

The Effectiveness of Robot-Assisted Origami Teaching: Verbal-Only vs. Verbal-and-Movement

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Abstract—Humans have always used gestures to convey ideas. The incorporation of gestures into speech makes it more memorable and easier to comprehend. This idea was explored in the study by implementing the same notion using robots. The motivation was to explore into the effectiveness of robot-assisted learning with accompanied gestures. A NAO robot was programmed to execute the instructions for building a paper fortune teller origami with gestures. The participants were segregated into two groups and their experience was noted in the form of feedback. This feedback was analysed to find that people have higher satisfaction and memory retention when the robot gives instructions with gestures as opposed to instructions without gestures.

Index Terms—assistive robots, human-robot interaction, effectiveness, teaching outcomes

Codes:https://github.com/OSiG43/HRI_study

I. INTRODUCTION

The integration of robotics into educational environments is a burgeoning field that aims to enhance learning experiences through interactive and engaging methods. One area of particular interest is the use of humanoid robots to assist in teaching complex tasks that traditionally rely heavily on human instruction and demonstration. Gestures, as a natural and fundamental aspect of human communication, play a crucial role in making instructions more comprehensible and memorable. This paper explores the impact of incorporating gestures in robot-assisted learning, specifically focusing on the task of teaching origami.

Origami, the traditional Japanese art of paper folding, requires precise and sequential instructions to achieve the desired outcomes. Learning such a skill often benefits from visual and gestural cues that complement verbal instructions. Previous studies have shown that gestures significantly aid in human learning by enhancing understanding and retention of information. However, there is a gap in research concerning the effectiveness of gestures when delivered by a robot instructor. This study aims to bridge this gap by investigating whether the

inclusion of gestures in robot-assisted teaching can improve the learning experience and outcomes.

The motivation for this research stems from the potential benefits that robot-assisted learning can offer. As educational tools, robots can provide consistent, repeatable, and tireless instruction, which can be particularly advantageous in settings where human instructors may not always be available. Moreover, understanding the role of gestures in robotic instruction can lead to the development of more effective teaching aids that can cater to a wide range of learning styles and needs.

To explore this, we programmed a NAO robot to teach participants how to make a paper fortune teller origami, with the robot providing instructions either with accompanying hand movements or without. The study was designed to compare the effectiveness of these two approaches by segregating participants into two groups and collecting feedback on their experiences. The primary research questions guiding this study are:

- Does the inclusion of gestures in robot-assisted teaching enhance learner satisfaction compared to verbal-only instructions?
- Do gestures provided by a robot improve memory retention of the instructional material?

By addressing these questions, this research aims to contribute valuable insights into the design and implementation of robotic teaching aids, ultimately enhancing the efficacy of robot-assisted learning across various educational contexts.

II. RELATED WORK

It is possible to find several studies in the literature that report on the use of robots on education. For example, Han et al. 2008 explored the educational impact of home robots on children's learning. They found that children had higher engagement and better information retention when interacting with the robot compared to traditional learning methods. This illustrates how interactive technologies such as robots can

enhance learning in environments such as home education or distance education

Shin and Kim 2007 investigated how students perceived the experience of learning with a robot. The study revealed that students generally had a positive attitude towards robot teaching and believed that robots could provide valuable assistance as study companions. This finding has important implications for our study as it is directly related to the evaluation of the acceptance and effectiveness of robots as interactive learning tools in user studies.

Kanda et al. 2004 conducted a field trial to investigate the role of robots as social partners and peer mentors. Their results show that children can form effective educational relationships with robots, which significantly help in the learning process, and it highlights the potential of robot interaction in complementing traditional educational Settings.

Tanaka and Matsuzoe 2012 Matsuzoe and Tanaka 2012 explore the educational benefits when children are responsible for teaching robots. They found that such inverse interactions not only improved children's understanding of the subject, but also enhanced empathy and teaching skills.

Saerbeck, Schut, Bartneck, and M. D. Janse 2010 studied how the level of expressive behavior of a robot affects learning outcomes in educational Settings. They demonstrated that more expressive robots are more effective at engaging students and enhancing learning.

III. METHODS

A. Research Question and Hypothesis

This study aims to evaluate the impact of explicative gestures on the effectiveness of teaching conducted by robots for manual jobs.

The following hypotheses are stated:

Hypothesis 1: Participants who receive teaching with oral instructions accompanied by gestures will demonstrate a better understanding of the instructions.

Hypothesis 2: Instructions accompanied by gestures are more memorable.

Hypothesis 3: Participants who receive teaching with oral instructions accompanied by gestures will report higher levels of satisfaction and engagement.

Hypothesis 1 will be tested by recording the number of times the participant needs to listen to each step and the number of times he needs to look at the paper's instructions. Hypothesis 2 and 3 will be tested using a post-experiment questionnaire using the User Experience Questionnaire Short (UEQ-S) completed with additional questions.

B. User Study Design

1) *Taught task:* The task the robot will teach must be manual, short enough to allow multiple repetitions within a 2-hour session, yet complex enough to challenge participants sufficiently to require reliance on the robot's instructions. Additionally, the task should be unfamiliar to most participants before the experiment.

Therefore, the chosen task is crafting a simple origami paper fortune teller. Although it is quite simple, it is complex enough to challenge participants, as each fold is crucial for completing the subsequent steps. The task involves approximately 12 steps and can be taught in about 10 minutes. The instructions may vary from the tutorial, with the selected tutorial being *12 Steps to Folding a Paper Fortune Teller - NEW Fine Arts Poster 2024*. The complete paper's instructions can be found in the appendix VII-A.

2) *Overview:* The study employed a between-subjects design where participants were randomly assigned to one of two groups to craft a paper fortune teller :

- **Group 1:** Origami instruction provided through oral instructions accompanied by gestures
- **Group 2:** Origami instruction provided through oral instructions only

Both groups can request the robot to repeat the step at the end of each instruction, and they can also consult the paper instructions (see Appendix VII-A) at any point during the learning process. The number of times participants request the robot to repeat a step and the number of times they refer to the paper instructions are recorded.

3) *Programming the Nao robot:* The NAO robot, developed by SoftBank Robotics, was selected as the robot platform for this experiment. Specifically designed for human-robot interaction, NAO is frequently employed in such experiments due to its versatility and ease of programming. Notably, NAO can be programmed using Choregraphe, a no-code programming interface, that allows the user to define scenarios with movement and speech easily and rapidly.

Two programs have been created:

Scenario 1: In this scenario, a custom movement is generated for each instruction, mimicking the required human movements to complete the step. This movement is executed immediately after the verbal instruction is given, aiming to illustrate the oral instructions visually. At the end of each step, the robot inquires whether the user would like to proceed to the next step or review the instructions once again. For convenience, the user can press the right foot of the robot to advance to the next step and the left foot to listen to the current step again. Both feet are clearly labelled to facilitate interaction.

Scenario 2: In this scenario, identical oral instructions to those in Scenario 1 are provided for each step, but no custom movements are executed. To ensure that users are placed under as similar conditions as possible, the robot is placed in the "Autonomous Life" motion mode (refer to the Autonomous Life chapter of *Choregraphe User Guide — Aldebaran 2.1.4.13 documentation 2024*) to engage in continuous, natural movements, preventing the robot from remaining immobile. Similarly to Scenario 1, the robot offers the option for the user to either listen to the step again or proceed to the next step.

The algorithm followed for each step is detailed in figure 1

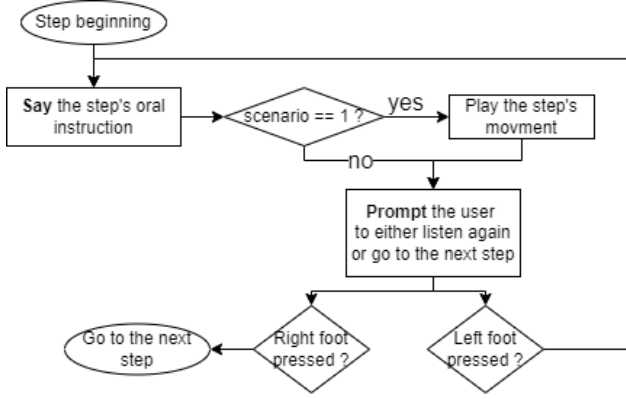


Fig. 1: Step algorithm

4) *Selection of the participants*: The ideal participant group for this study consists of individuals who do not regularly practice origami. This group should be diverse in terms of background and gender, aiming to form a representative sample of the UK population. However, due to time constraints for executing the experiment, participants will mainly be recruited from individuals working or studying at the Frenchay Campus of the University of the West of England.

C. User Study Procedure

1) *User welcome*: The user is greeted by a team member who requests them to read the information sheet (appendix VII-I) and the privacy notice (appendix VII-H). If the user consents to participate in the experiment, they are asked to complete the consent form (Appendix VII-H). To prevent bias in the experiment, users are briefed using an information sheet. This sheet outlines that users will learn to craft a simple origami and that they can ask the robot to explain a step again. Additionally, users are informed that they can refer to the paper instructions if they feel they do not understand a step with the robot's instruction.

2) *Pre-Experiment*: The user is given a unique ID to identify its answers across the different questionnaires and metrics recorded. Then, he is asked to scan a QR code and complete the online questionnaire 1 (appendix VII-C), which collects information about the user to determine their background.

3) *The Experiment*: The Experiment is designed in a way in which the robot automatically interacts with the user. Therefore, once the experiment is started, one team members carefully look at the participant to record the number of times the user has listened to a step (minimum is once per step meaning twelve) and the number of times the user has watched the paper's instructions.

4) *Post-Experiment*: When the experiment is over, the participant is asked to complete the post-experiment questionnaire (appendix VII-D) that aims to evaluate:

- Their confidence about their ability to complete the origami tasks independently
- The difficulty of the teaching process: how pleasant was the learning process.

- Understanding of robot guidance : How well the user understands the instructions given by the robot
- The helpfulness of robot teaching : How helpful the robot has been in teaching, guiding and interacting with the user

D. Dependent Measures

1) *Objective variables*: To assess the participants' understanding of the robots' instructions, the number of times they listen to the instructions and the number of times they look at the paper instructions are recorded. These figures are recorded individually for each stage during the experiments, but because we are only interested in overall understanding and not in the variation in difficulty between stages, only the summed values will be analysed.

Designation	measured metrics
M1	Total number of listens
M2	Total number of looks at the paper instructions

TABLE I: Objective measures

2) *Subjective variable*: In order to retrieve the demographic information of the participant group, the pre-experiment questionnaire (appendix VII-C) prompts the user for their age range, gender, and experience with origami and with assistive robots in general and the subjective evaluation of the origami teaching using robots.

To complete the assessment of the participant's understanding of the robots' instructions provided by the objective metrics, questions Q2 and Q3 of the post-experiment questionnaire (appendix VII-D1) are used.

To assess the memorability of the instructions, the original plan was to have the user create the same origami independently after the learning process. However, due to time constraints, the experiment only includes question Q1, which asks about the user's confidence in completing the origami on their own.

Finally, in order to quantify the quality of the user experience, the UEQ-S (User Experience Questionnaire Short) is used. This questionnaire is a short version of the User Experience Questionnaire developed by Laugwitz et al Laugwitz, Held, and Schrepp 2008 aiming to provide a standardized measurement of the user experience using various indicators. The short version used in this study has been designed and tested by Shrepp et al Schrepp, Hinderks, and Thomaschewski 2017a and allows to evaluate user experience using two subscales presented in figure 2: Pragmatic and Hedonic quality.

The UEQ-S questionnaire can be completed by question Q2 and Q3 both relating to pragmatic quality.

E. Result analysis

To analyse the results, the normality of each metrics will be tested using the Shapiro-Wilk test **SHAPIRO1965** and the appropriate association test, Student t-test **student'probable'1908** for normally distributed metrics or the Mann-Whitney U test **noauthor'mannwhitney'nodate** for non normally distributed metrics will be used to determine if there is a significant difference between the two groups.

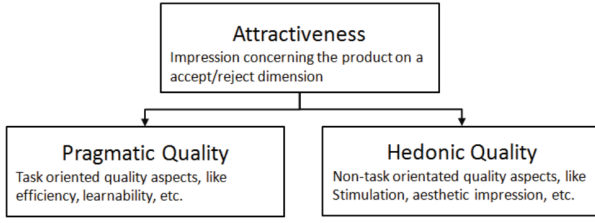


Fig. 2: Content of UEQ-S quality metrics (Rauschenberger et al. 2013)

IV. RESULTS

A. Participants

The recruited set of participants consists of 10 people working or studying on the Frenchay campus with 6 of them being students of the Human Robot Interaction tutorial group. Due to the lack of time to run the experiment, we were unable to include more participants with different backgrounds. The characteristics of the set of participants are given by questionnaire 1 ([[ADD LINK TO METHOD SECTION](#)]) and the results are listed in Table II

Attribute	Group 1	Group 2
male	100%	60%
female	0%	40%
Age: 18-25	60%	80%
Age: 26-30	20%	20%
Age: 41-50	20%	0%
Practice origami once a year	20%	40%
Practice origami less than once a year	80%	60%

TABLE II: Distribution of participants

B. Results visualization

To explore if there are differences between the group with motion (Group 1) and the group with speech only (Group 2), bar plots with whiskers showing the average answer and the standard deviation for each group have been plotted for the objective metrics (number of listenings and number of paper instructions watched) on figure 3, the answers of the additional questions (appendix VII-D1) on figure 4 and answers of the short User Experiment Questionnaire (appendix VII-D2) on figure 5.

Additionally, histograms with both groups have been plotted for each question of the UEQS and the additional questions, as an example, figure 6 shows the histogram for question 1: ”If you had to do it again on your own, how far do you think you’d be able to go? (term of steps)” (see appendix VII-D1)

C. Normality test

To decide which test of association can be used for each question, a normality test should be run. Therefore, the Shapiro-Wilk normality test is applied to all the answers the number of listens and the number of paper instruction watches. Figure 7 presents the obtained p-scores. M1 refers to the number of listens and M2 to the number of paper watches.

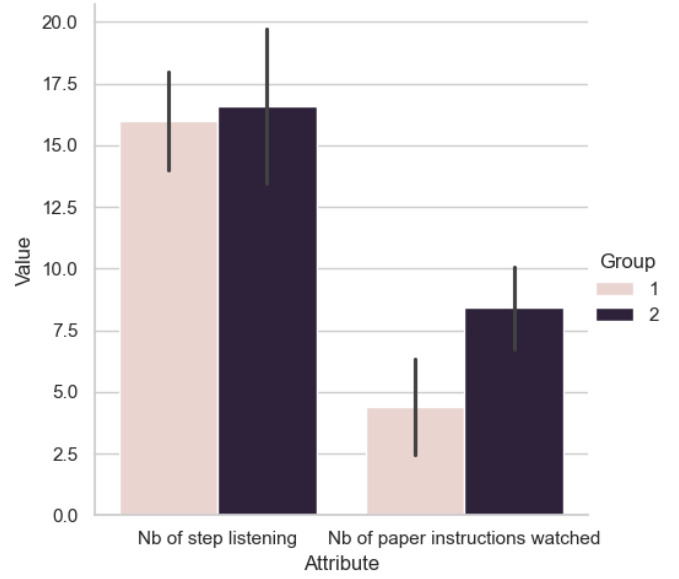


Fig. 3: Objective variables results

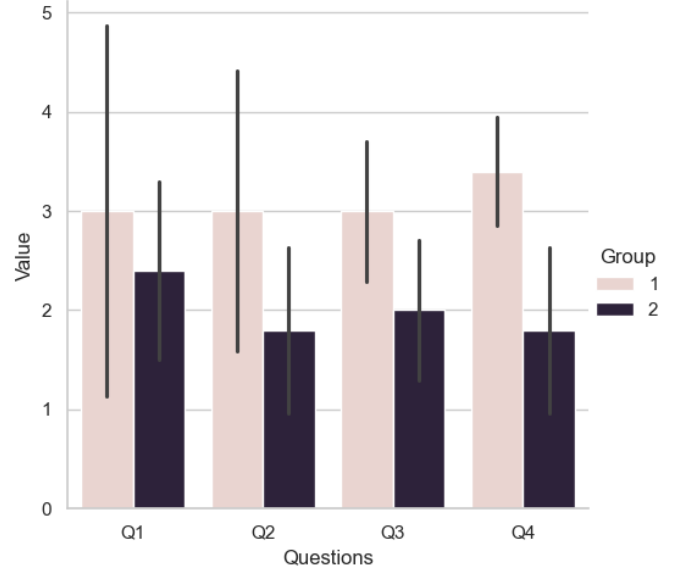


Fig. 4: Additional questions results

Q1 to Q4 still refer to the additional questions and U1 to U8 referees to UEQ-S answers. The dashed line drawn at $p=0.05$ represents the common threshold value to reject the null hypothesis meaning for this test that a value under $p=0.05$ means that the answers for this question do not follow a normal distribution.

D. Test of association

Considering a threshold of $p=0.05$, the normality test performed in section IV-C shows that the results for M1, Q1, Q3, U1, U2, U3 and U5 can be considered normal while the result for M2, Q2, Q4, U4, U6, U7 and U8 cannot be considered normal. Therefore two different association tests were run,

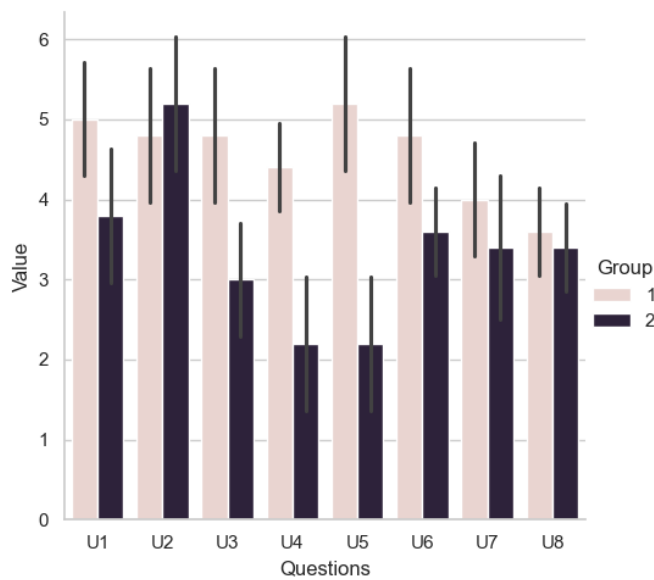


Fig. 5: UEQ-S results

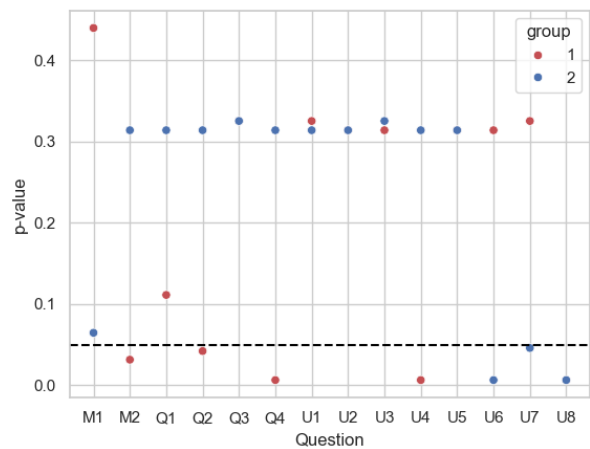


Fig. 7: Normality p-score using the Shapiro-Wilk test

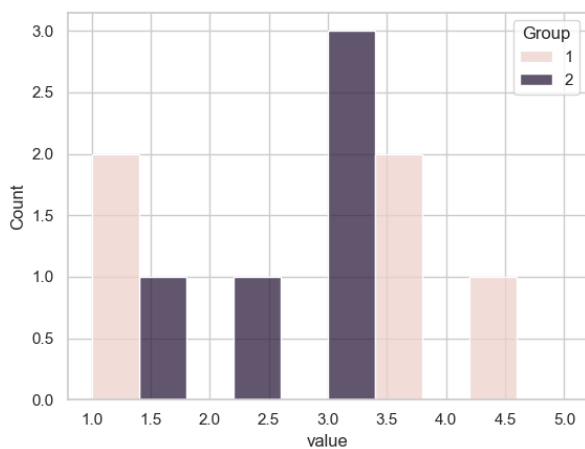


Fig. 6: Distribution of answers for question Q1

the normal result was tested using the t-test and the other was tested using the Mann Whitney test.

Figure 8 shows a summary of the p-score for each question, the calculation method being the one described before (either t-test or Mann Whitney test)

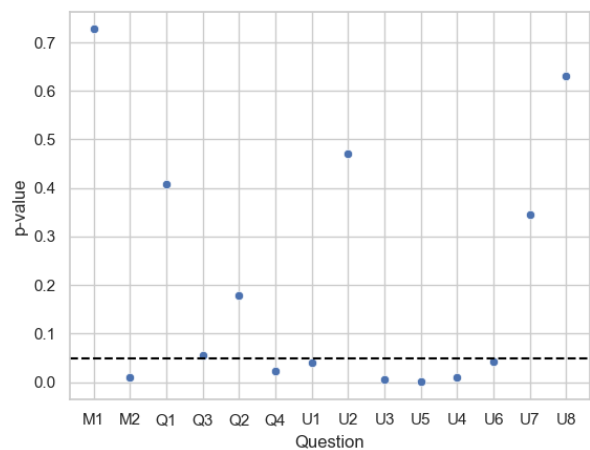


Fig. 8: Result of the association test

E. UEQ-S results

Additionally, the calculation tool provided with the UEQ-S gives an overall analysis for the UEQ-S answer calculating a score for the pragmatic and hedonic quality and showing a bar plot with whiskers and a t-test to assess the statistical differences between the two groups.

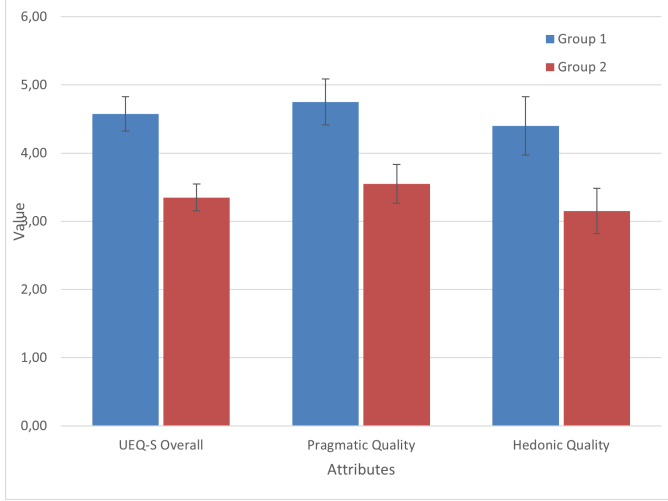


Fig. 9: Result of the UEQ-S quality metrics

Figure 9 shows the results of the pragmatic and hedonic quality for both groups calculated using the calculation sheet provided by Shrepp et al on their website [hinderks' user'2024](#).

Additionally, table III show the result of the t-test:

Attributes	p-score	Difference
UEQ-S Overall	0,0001	Significant Difference
Pragmatic Quality	0,0009	Significant Difference
Hedonic Quality	0,0022	Significant Difference

TABLE III: UEQ-S t-test results

V. DISCUSSION

The study conducted involved taking the opinion of the users on the robot's process. The results of the study were analyzed. The survey was conducted with 10 students. They were divided into two groups.

- Group 1 was subject to the instructions of the Origami process with the NAO robot speaking the instructions accompanied with a demonstration by gestures and movement.
- Group 2 was subject to instructions only, and was not demonstrated the procedure by the paper.

They were also supplied with a paper that had the instructions printed on them with relevant diagrams. Their experience during the survey was recorded based on two parameters:

- Number of times the participants followed the steps. This was used to keep a track of the number of times the participants from both the groups watched each step. This kept a count of the number of times the steps were revisited.

- Number of times instructions printed on paper were referenced to. This was used to keep a track of the number of times both the groups of participants referenced the instructions printed on the paper.

The results that were recorded were visualized in the form of histograms, that illustrated the differences between the groups with motion and without motion. It was observed that Group 1 showed a better understanding of the Origami process. The number of times they had to go back to the previous step were lesser and the number of times they had to refer to the sheet with the printed steps were also less.

Statistical methods were applied to interpret the data. Normality test was carried out and the p-score was calculated. The Shapiro-Wilk Normality test was carried out to decide which test of association could be used. The p-value is a measure of the strength of evidence against the null hypothesis in a hypothesis test. It indicates the probability of observing the test statistic (or more extreme results) if the null hypothesis is true.

It is possible to consider the results for M1, Q1, Q3, U1, U2, U3, and U5 to be normal, but the results for M2, Q2, Q4, U4, U6, U7, and U8 cannot be typical. As a result, two separate association tests were conducted: one used the Mann Whitney test to evaluate the normal result, and the other used the t-test.

This study offers more evidence that robot movements not only make interactions more engaging but also work better when teaching complicated skills like origami, in line with the findings of Saerbeck, Schut, Bartneck, and M. Janse 2010 study demonstrating the potential of expressive robotics to improve learning results. In contrast, the current work directly adds to this, whereas research like Ma et al. 2015 concentrated on involvement without explicitly evaluating the impact of task completion.

The study conducted creates an understanding of the effectiveness of the integration of robots in educating people. The study illustrates how a robot with gestures can be an effective tool in education and engaging interaction. This encourages further research in exploring how lifelike the interaction can be created so that the users can follow the procedures more easily.

The study was not without its own limitations. Head tracking was implemented in the NAO robot. The NAO Robot is able to track targets using the Tracker module. However, it is done assuming average target size, so it is not very accurate. The head tracking worked smoothly when the robot was interacting with the subject, however when it went to the function where it demonstrated the origami steps through gestures it lost the subject, and did not track their head. This resulted in some jerky movements that were very robot-like. This was a step back in bridging the gap in making Human-Robot Interaction lifelike.

The study was conducted with participants that had some level of experience working with robots. Hence the study proceeded without major obstacles. However, due to the time constraints, the survey could not be more diverse, including

more participants from different backgrounds. The functioning of the robot went smoothly except the head tracking part. The statistical methods applied were adequate in measuring the distribution of the results over the participants and the groups they were categorized into. An improvement on the survey procedure would bring forward more participants, giving a wider pool of test subjects and could have resulted into a more extensive survey. Also using the UEQ Questionnaire instead of UEQ-S would have been a better choice. This is because it is a far more comprehensive approach and is more reliable in conducting surveys where the outcome of the survey analyses the user experience.

VI. CONCLUSION

The study was conducted to explore the Human Robot Integration in terms of teaching to assess the feasibility of the integration of such a robot into the teaching environment. The overall observation from the survey was found to be quite positive, showing results that indicate towards the success of a fluent Human Robot Interaction in the teaching environment. It also illustrates that people when assisted with gestures that supplement audio instructions report higher levels of satisfaction and a better understanding and memory retention of the instructions.

The study shows that in the field of Human Robot Interaction, interactions on seemingly complex tasks like the folding of intricate origami can be made easier by supplementing the interaction with appropriate gestures. In a broader perspective, incorporation of gestures makes a process more likely to be understood by the user and also leaves a more memorable instruction in the users' mind. These study results can be considered for further study and would be useful for people exploring the effectiveness of gesture and movement in teaching by robots. The research has ample scope of further improvement by making the hand gestures smoother and the head tracking more well integrated into the system and improving upon the voice modulation to make the interaction more lifelike and natural. The survey conducted can be replicated on a wider and more diverse participant base with a more extensive analysis of the results.

Overall the study conducted was able to prove the hypotheses to be true and was able to bring out new insights into the implementation of robots in teaching environments.

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VII. APPENDIX

A. Paper's instructions

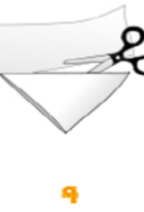
Step number	Image	Step number	Image
Step 1		Step 7	
Step 2		Step 8	
Step 3		Step 9	
Step 4		Step 10	
Step 5		Step 11	
Step 6		Step 12	

Fig. 10: Paper's instruction for the fortune teller

B. Instructions for the Origami: Audio instructions

Step Number	Instruction
Step 1	Take one sheet of paper
Step 2	Firstly, the goal is to make a perfect square. To do it, take your sheet of paper and fold it across diagonally lining up the two sides
Step 3	Fold across the other diagonal and line up the corner with the previous fold
Step 4	Cut the excess part of the paper
Step 5	Great! Now you can unfold the paper. You should get a perfect square with visible folds along the diagonals
Step 6	Fold the corner of the paper to meet the centre
Step 7	Repeat this for the other three corners
Step 8	Flip the paper square on its blank side
Step 9	Fold each corner of the square to its centre. You should obtain a smaller square
Step 10	First Fold the square lengthwise. Then fold the square again on itself
Step 11	Smoothen the folds so they are clear
Step 12	Place your Thumb and index fingers inside the pockets in the paper. Open your fingers to get the final form

TABLE IV: Audio Instructions

C. Pre-Experiment Questions

1) *Addition Questions:* These pre-experiment questions were developed to align more closely with our research focus and hypothesis, complementing the UEQ-S questionnaire in pre-experimental section. In this section, Participants will be asked to complete a questionnaire assessing their origami skills, learning preferences, and other background information associated with the content of experiments.

1) What is your age range?

- ☐ Under 18
- ☐ 18 25
- ☐ 26 30
- ☐ 31 40
- ☐ 41 50
- ☐ 51 60
- ☐ Over 60

2) What is your gender?

- ☐ Male
- ☐ Female

3) Did you have any experience with origami before participating in this study?

- ☐ Less
- ☐ Once a year or more
- ☐ Once a month or more
- ☐ Once a week or more
- ☐ More

4) Do you have any experience with assistive robots?

- ☐ Less
- ☐ Once a year or more
- ☐ Once a month or more
- ☐ Once a week or more
- ☐ More

5) Your subjective evaluation of the NAO robot teaching origami? (1 means the best, 5 means the worst)

- ☐ Excellent
- ☐ Good
- ☐ Average
- ☐ Poor
- ☐ Very Poor

D. Post-Experiment Questions

1) *Additional questions:* These post-experiment questions were proposed to make the study more specifically to our research focus and hypothesis, supplementing the UEQ-S questionnaire in post-experimental section. In this part, participants will be asked to provide feedback and comments on instruction methods and their learning experience.

1) Q1: If you had to do it again on your own, how far do you think you'd be able to go? (term of steps)

- ☐ I do not feel able to complete any step
- ☐ Step 1-4
- ☐ Step 5-8
- ☐ Step 9-11
- ☐ I feel able to complete all steps

2) Q2: How would you rate the overall difficulty of the learning process?

- ☐ Very difficult
- ☐ Somewhat difficult
- ☐ Moderate
- ☐ Somewhat easy
- ☐ Very easy

3) Q3: How well could you understand the robot's instructions during the learning process?

- ☐ Could not understand at all
- ☐ Very difficult to understand
- ☐ Somewhat difficult to understand
- ☐ Mostly understood
- ☐ Completely understood

4) Q4: Do you think the robot's guidance helps complete the origami?

- ☐ Not helpful at all
- ☐ Unhelpful
- ☐ Not very helpful
- ☐ Helpful
- ☐ Extremely helpful

2) *User Experience Questionnaire Short:* .

obstructive	o o o o o o o	supportive
complicated	o o o o o o o	easy
inefficient	o o o o o o o	efficient
confusing	o o o o o o o	clear
boring	o o o o o o o	exciting
not interesting	o o o o o o o	interesting
conventional	o o o o o o o	inventive
usual	o o o o o o o	leading edge

Fig. 11: UEQ-S Questionnaire (Schrepp, Hinderks, and Thomaschewski 2017b)

E. Questions histograms

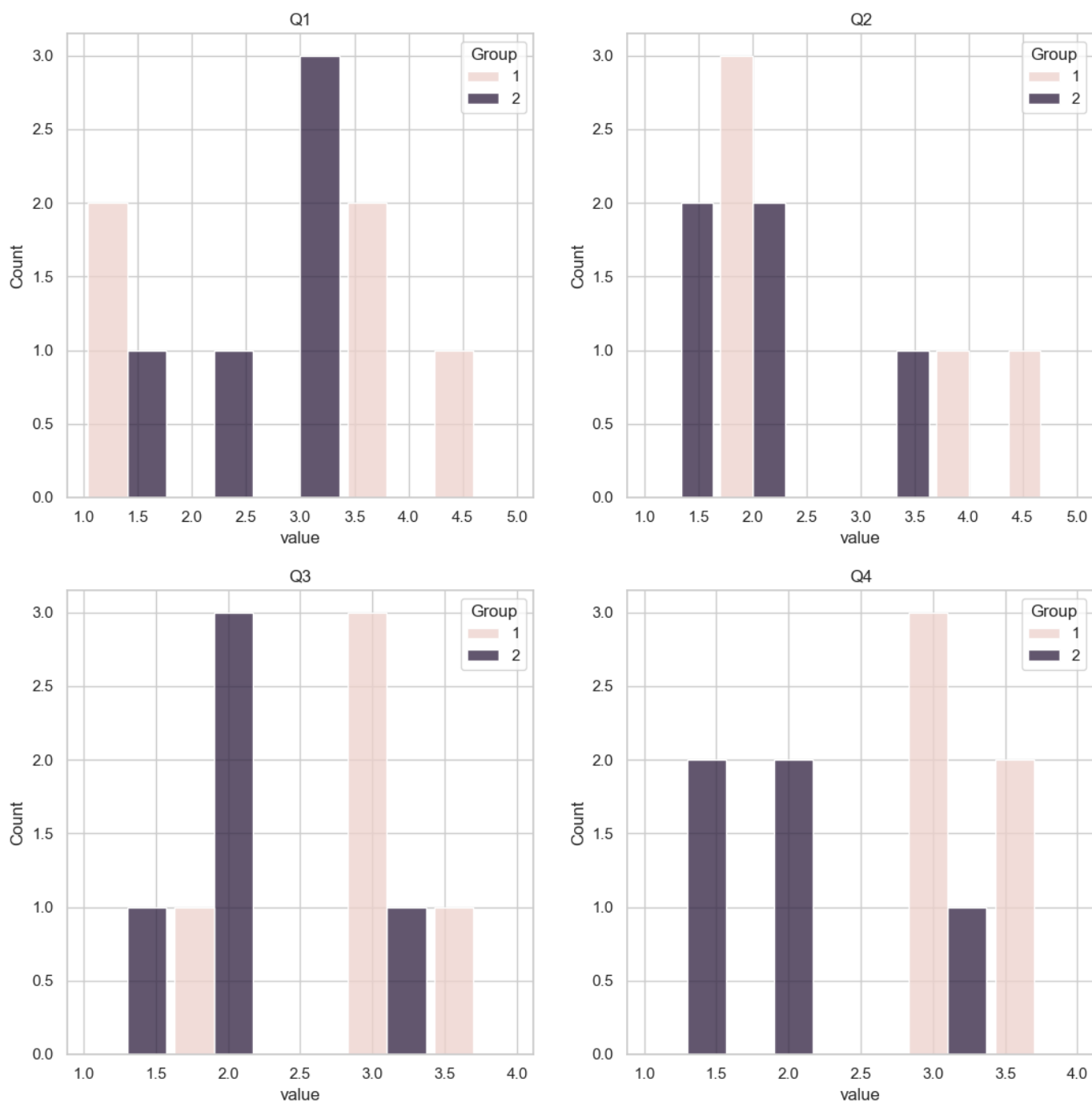


Fig. 12: Additional questions histogram

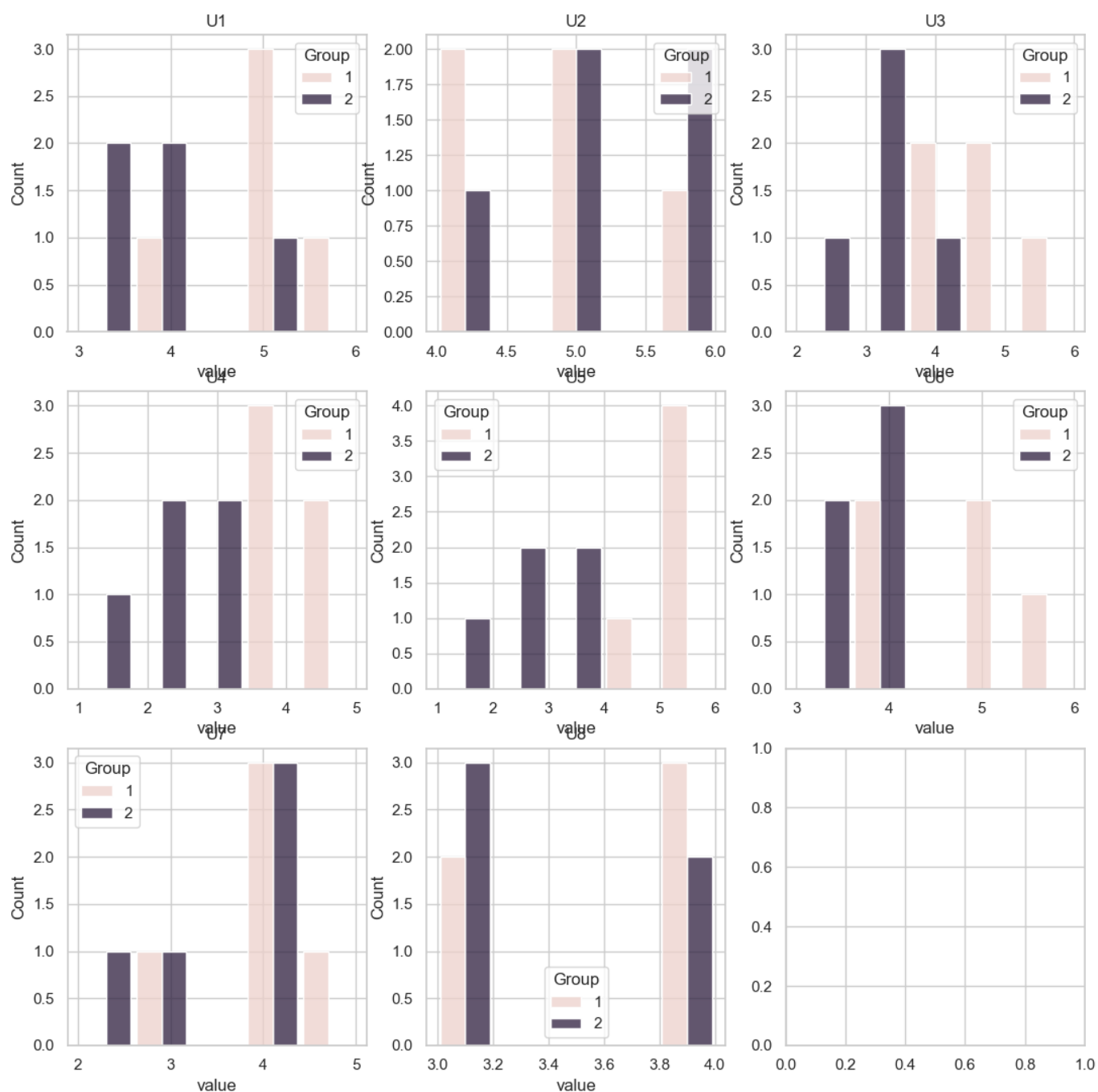


Fig. 13: UEQ-S histogram

F. Informed Consent Form

Informed Consent Form.pdf



Faculty of Environment & Technology
Faculty Research Ethics Committee (FREC)

Ethical Review Checklist for Undergraduate and Postgraduate Modules

Staff and PG research students must not use this form, but should instead, if appropriate, submit a full application for ethical approval to the Faculty Research Ethics Committee (FREC).

Please provide project details and complete the checklist below.

Project Details:

Module name	Human-robot interaction
Module code	UFMFHP-15-M
Module leader	Dr Paul Bremner
Project Supervisor	Dr Paul Bremner
Proposed project title	Exploring the Effectiveness of Verbal-Only vs. Verbal-and-Movement in Robot-Assisted Origami Teaching

Applicant Details:

Name of Student	Chuanbeibei Shi; Kai Yang; Madhurya Mozumder; Simon Galand
Student Number	23080504 23080468 23080485 23080546
Student's email address	Chuanbeibei2.shi@live.uwe.ac.uk ; Kai3.Yang@live.uwe.ac.uk ; Madhurya2.Mozumder@live.uwe.ac.uk ; simon2.galand@uwe.ac.uk .

CHECKLIST QUESTIONS		Yes/No	Explanation
1.	Does the proposed project involve human tissue, human participants, animals, environmental damage, or the NHS.	Yes	The project focuses on comparing two teaching methods: verbal-only instruction and instruction that incorporates both verbal guidance and movements provided by our robot. Participants will be engaged in hands-on origami activities guided by the robot using these two methods.

H. Privacy Notice

Privacy notice.pdf



Study Information Sheet

Study Title: Exploring the Effectiveness of Robot-Assisted Origami Teaching

PLEASE READ THIS SHEET IN ITS ENTIRETY

You are invited to take part in research taking place at the University of the West of England, Bristol. It is carried out as assignment for module UFMFHP-15-M Human-Robot Interaction. Before you decide whether to take part, it is important for you to understand why the study is being done and what it will involve. Please read the following information carefully and if you have any queries or would like more information please contact *Simon Galand*, Madhurya Mozumder, Chuanbeibei Shi, Kai Yang, Faculty of Environment and Technology, Bristol Robotics Laboratory, University of the West of England, Bristol.

simon2.galand@uwe.ac.uk.

Madhurya2.Mozumder@live.uwe.ac.uk

chuanbeibei2.shi@live.uwe.ac.uk

Kai3.Yang@live.uwe.ac.uk

Who is organising the research?

The project is led by *Simon Galand*, Madhurya Mozumder, Chuanbeibei Shi, Kai Yang, University of the West of England. Manuel Giuliani is the supervisor for this research. Please find their details at the end of this document.

What is the aim of the research?

The overall aim of the research is to investigate the impact of instruction modalities on learning outcomes in robot-assisted origami teaching

Why have I been invited to take part?

We are recruiting participants who are already working at the University of the West of England. People who are a person in need of assistance, deaf, blind have a lack of manual flexibility or are child cannot take part to this study.

Do I have to take part?

You do not have to take part in this research. It is up to you to decide whether or not you want to be involved. If you do decide to take part, you will be given a copy of this information sheet to keep and will be asked to sign a consent form. If you do decide to take part, you are free to stop and withdraw from the study at any time during the study without giving a reason.

Label and QR.pdf

VIII. CODE FOR STATISTICS

A. Data Analysis

```
1 data = pd.read_csv('merged_results.csv')
2
3 data.head()
4
5 # Create a new data set with the column: 'id', '
  group', 'value', 'attribute'
6 # where 'attribute' is the name of the column (
  Listening, Paper, Q1, Q2,Q3,Q4) and 'value' is
  the response to the question
7
8 dataMetrics = data.melt(id_vars=['Id', 'Group'],
  value_vars=['Listening', 'Paper'], var_name='
  attribute', value_name='value')
9 dataQuestionnaire = data.melt(id_vars=['Id', 'Group']
  , value_vars=['Q1', 'Q2', 'Q3', 'Q4'],
  var_name='attribute', value_name='value')
10 dataUEQS = data.melt(id_vars=['Id', 'Group'],
  value_vars=['U1','U2','U3','U4', 'U5', 'U6', 'U7'
  ', 'U8'], var_name='attribute', value_name='
  value')
11
12 #Renaming attribute of dataMetrics to 'Nb of step
  listening' and 'Nb of paper instructions watched'
13 dataMetrics['attribute'] = dataMetrics['attribute'].
  replace({'Listening': 'Nb_of_step_listening', '
  Paper': 'Nb_of_paper_instructions_watched'})
14
15 data.describe( include='all')
16
17 # Plot bar chart of mean and standard deviation with
  all question responses for each group
18 import seaborn as sns
19 import matplotlib.pyplot as plt
20
21 sns.set_theme(style="whitegrid")
22 sns.color_palette("hls", 8)
23
24 f = sns.catplot(x="attribute", y="value", hue="Group"
  , data=dataMetrics, kind="bar", errorbar='sd')
25 #set axis labels
26 f.set_axis_labels("Attribute", "Value")
27 #Save the figure
28 f.savefig('metrics.png')
29
30 f = sns.catplot(x="attribute", y="value", hue="Group"
  , data=dataQuestionnaire, kind="bar", errorbar=
  'sd')
31 f.set_axis_labels("Questions", "Value")
32 #Save the figure
33 f.savefig('questionnaire.png')
34
35 f = sns.catplot(x="attribute", y="value", hue="Group"
  , data=dataUEQS, kind="bar", errorbar='sd')
36 f.set_axis_labels("Questions", "Value")
37 f.savefig('ueqs.png')
38
39
40 #Plot histogram for the question responses (Q1, Q2,
  Q3, Q4 and U1, U2, U3, U4, U5, U6, U7, U8)
41 fig, axs = plt.subplots(2, 2, figsize=(13, 13))
42 #Plot on each subplot the histogram of the question
  responses for both groups
43 #For example, axs[0,0] is the first subplot, with
  the histogram of the question responses Q1 for
  Group 1 and Group 2
44 for i, question in enumerate(['Q1', 'Q2', 'Q3', 'Q4'
  ]):
45     sns.histplot(dataQuestionnaire[dataQuestionnaire
  ['attribute'] == question], x='value', hue='
  Group', multiple="dodge", ax=axs[i//2, i%2],
  shrink=.9)
  axs[i//2, i%2].set_title(question)
46
47
48 #save
49 plt.savefig('questionnaire_histogram.png')
50
51 #Plot histogram for the question responses ( U1, U2,
  U3, U4, U5, U6, U7, U8)
52 fig, axs = plt.subplots(3, 3, figsize=(13, 13))
53 #Plot on each subplot the histogram of the question
  responses for both groups
54 #For example, axs[0,0] is the first subplot, with
  the histogram of the question responses Q1 for
  Group 1 and Group 2
55 axs = axs.flatten()
56 for i, question in enumerate(['U1', 'U2', 'U3', 'U4'
  , 'U5', 'U6', 'U7', 'U8']):
57     sns.histplot(dataUEQS[dataUEQS['attribute'] ==
  question], x='value', hue='Group', multiple=
  "dodge", ax=axs[i], shrink=.9)
  axs[i].set_title(question)
58
59
60 #save
61 plt.savefig('ueqs_histogram.png')
62
63 #plot the histo for Q1
64 f = sns.histplot(dataQuestionnaire[dataQuestionnaire
  ['attribute'] == 'Q1'], x='value', hue='Group',
  multiple="dodge")
65 #Save the figure
66 f.figure.savefig('Q1_histo.png')
67
68
69 #Normality test with Shapiro-Wilk
70 from scipy.stats import shapiro
71
72 #create dataset normality with 'group' 'attribute'
  and 'normality_p'
73 normalityList = []
74
75 for i, metric in enumerate(['Nb_of_step_listening',
  'Nb_of_paper_instructions_watched']):
76     for group in [1,2]:
77         responses = dataMetrics[(dataMetrics['
  attribute'] == metric) & (dataMetrics['
  Group'] == group)][['value']]
78         stat, p = shapiro(responses)
79         normalityList.append({'group': group, '
  attribute': ["M1","M2"][i], 'normality_p'
  ': p})
80
81 #For each question, test the normality of the
  responses for both groups
82 for question in ['Q1', 'Q2', 'Q3', 'Q4']:
83     for group in [1,2]:
84         #Get the responses for the question and
  group
85         responses = dataQuestionnaire[(
  dataQuestionnaire['attribute'] ==
  question) & (dataQuestionnaire['Group']
  == group)][['value']]
86         #Test the normality of the responses
87         stat, p = shapiro(responses)
88         #Add the result to the dataNormality
  dataframe
89         normalityList.append({'group': group, '
  attribute': question, 'normality_p': p})
90
91
92 #Same for UEQS
93 for question in ['U1', 'U2', 'U3', 'U4', 'U5', 'U6',
  'U7', 'U8']:
94     for group in [1,2]:
```

```

95     responses = dataUEQS[(dataUEQS['attribute'] == question) & (dataUEQS['Group'] == group)][['value']]
96     stat, p = shapiro(responses)
97     normalityList.append({'group': group, 'attribute': question, 'normality_p': p})
98
99
100 dataNormality = pd.DataFrame(normalityList)
101
102 #Plot the normality test results
103 sns.set_theme(style="whitegrid")
104 f = sns.scatterplot(data=dataNormality, x='attribute', y='normality_p', hue='group', palette=['r', 'b'])
105 #Plot a horizontal line at y=0.05
106 f.axhline(0.05, color='black', linestyle='--')
107 #Set the axis labels
108 f.set(xlabel='Question', ylabel='p-value')
109
110 #Save the figure
111 f.figure.savefig('normality_result.png')
112
113 # Perform the t-test for each question
114 from scipy.stats import mannwhitneyu #Perform the t-test for each question
115 from scipy.stats import ttest_ind
116
117 #Create a dataframe to store the t-test results
118 testList = []
119
120 #perform t-test for the metrics
121 rep1 = dataMetrics[dataMetrics['Group'] == 1]
122 rep2 = dataMetrics[dataMetrics['Group'] == 2]
123
124 #t test for number of step listening
125 stat, p = ttest_ind(rep1[rep1['attribute'] == 'Nb_of_step_listening'], rep2[rep2['attribute'] == 'Nb_of_step_listening'])
126 testList.append({'attribute': 'M1', 't_stat': stat, 'p_value': p})
127
128 # U test for number of paper instructions watched
129 stat, p = mannwhitneyu(rep1[rep1['attribute'] == 'Nb_of_paper_instructions_watched'], rep2[rep2['attribute'] == 'Nb_of_paper_instructions_watched'])
130 testList.append({'attribute': 'M2', 't_stat': stat, 'p_value': p})
131
132 # T test for normally distributed data
133 #For each question, perform the t-test for the responses of both groups
134 for question in ['Q1', 'Q3']:
135     #Get the responses for the question for both groups
136     responses1 = dataQuestionnaire[(dataQuestionnaire['attribute'] == question) & (dataQuestionnaire['Group'] == 1)][['value']]
137     responses2 = dataQuestionnaire[(dataQuestionnaire['attribute'] == question) & (dataQuestionnaire['Group'] == 2)][['value']]
138     #Perform the t-test
139     stat, p = ttest_ind(responses1, responses2)
140     #Add the results to the ttestList dataframe
141     testList.append({'attribute': question, 't_stat': stat, 'p_value': p})
142
143 # Mann-Whitney U test for non-normally distributed data
144 for question in ['Q2', 'Q4']:
145     responses1 = dataQuestionnaire[(dataQuestionnaire['attribute'] == question) & (dataQuestionnaire['Group'] == 1)][['value']]
146     responses2 = dataQuestionnaire[(dataQuestionnaire['attribute'] == question) & (dataQuestionnaire['Group'] == 2)][['value']]
147     stat, p = mannwhitneyu(responses1, responses2)
148     testList.append({'attribute': question, 't_stat': stat, 'p_value': p})
149
150 #Same for UEQS
151 for question in ['U1', 'U2', 'U3', 'U5']:
152     responses1 = dataUEQS[(dataUEQS['attribute'] == question) & (dataUEQS['Group'] == 1)][['value']]
153     responses2 = dataUEQS[(dataUEQS['attribute'] == question) & (dataUEQS['Group'] == 2)][['value']]
154     stat, p = ttest_ind(responses1, responses2)
155     testList.append({'attribute': question, 't_stat': stat, 'p_value': p})
156
157 for question in ['U4', 'U6', 'U7', 'U8']:
158     responses1 = dataUEQS[(dataUEQS['attribute'] == question) & (dataUEQS['Group'] == 1)][['value']]
159     responses2 = dataUEQS[(dataUEQS['attribute'] == question) & (dataUEQS['Group'] == 2)][['value']]
160     stat, p = mannwhitneyu(responses1, responses2)
161     testList.append({'attribute': question, 't_stat': stat, 'p_value': p})
162
163 dataTTest = pd.DataFrame(testList)
164
165 #Plot the t-test results
166 sns.set_theme(style="whitegrid")
167 f = sns.scatterplot(data=dataTTest, x='attribute', y='p_value')
168 #Set the axis labels
169 f.set(xlabel='Question', ylabel='p-value')
170 #Plot a horizontal line at y=0.05
171 f.axhline(0.05, color='black', linestyle='--')
172
173 #Save the figure
174 f.figure.savefig('association_result.png')
175
176 #Compute t test for the metrics
177 ttestList = []
178
179 for metric in ['Nb_of_step_listening', 'Nb_of_paper_instructions_watched']:
180     responses1 = dataMetrics[(dataMetrics['attribute'] == metric) & (dataMetrics['Group'] == 1)][['value']]
181     responses2 = dataMetrics[(dataMetrics['attribute'] == metric) & (dataMetrics['Group'] == 2)][['value']]
182     stat, p = ttest_ind(responses1, responses2)
183     ttestList.append({'attribute': metric, 't_stat': stat, 'p_value': p})
184
185 dataTTestMetrics = pd.DataFrame(ttestList)
186
187 #Plot the t-test results
188 sns.set_theme(style="whitegrid")

```

```
195 f = sns.scatterplot(data=dataTtestMetrics, x='
    attribute', y='p_value')
196
197 #Plot a horizontal line at y=0.05
198 f.axhline(0.05, color='black', linestyle='--')
```